

IVALO, Finland

WMO: 028070

Lat: 68.608N

Lon: 27.414E

Elev: 147

StdP: 99.58

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.7	(c) -29.0	(d) -35.9	(e) 0.1	(f) -32.6	(g) -31.9	(h) 0.2	(i) -28.8	(j) 9.9	(k) -5.1	(l) 8.7	(m) -4.5	(n) 1.2	(o) 240	(p) 0.727

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	24.8	16.1	22.4	15.1	20.3	14.2	17.5	22.5	16.2	20.7	15.0	19.1	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.2	19.2	14.2	10.3	17.9	13.1	9.6	16.7	49.7	22.3	46.0	20.9	42.4	19.1	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.4	(b) 7.3	(c) 6.5	(e) -35.6	(f) 27.8	(g) 4.0	(h) 2.2	(i) -38.5	(j) 29.3	(k) -40.9	(l) 30.6	(m) -43.1	(n) 31.8	(o) -46.1	(p) 33.4		
(a) 8.4	(b) 7.3	(c) 6.5	(e) -35.7	(f) 18.8	(g) 3.8	(h) 1.4	(i) -38.4	(j) 19.8	(k) -40.7	(l) 20.7	(m) -42.8	(n) 21.5	(o) -45.6	(p) 22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-12.4	-12.1	-7.5	-1.5	4.6	10.5	14.2	11.9	7.0	0.3	-6.3	-9.6	(6)
(7)		DBStd	10.77	7.96	8.09	6.12	4.73	4.22	4.17	3.81	3.55	3.64	4.65	6.25	7.54	(7)
(8)		HDD10.0	3951	694	619	543	345	178	43	7	19	104	301	488	609	(8)
(9)		HDD18.3	6711	953	852	802	595	427	238	136	203	341	559	738	867	(9)
(10)		CDD10.0	295	0	0	0	0	9	58	138	77	13	0	0	0	(10)
(11)		CDD18.3	14	0	0	0	0	0	2	9	3	0	0	0	0	(11)
(12)		CDH23.3	134	0	0	0	0	4	22	84	25	0	0	0	0	(12)
(13)		CDH26.7	18	0	0	0	0	0	3	12	3	0	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.5	2.7	3.0	2.8	3.1	3.1	2.8	2.4	2.6	2.7	2.5	2.8	(14)
(15)	Precipitation	PrecAvg	393	20	16	19	18	27	44	64	60	44	35	26	21	(15)
(16)		PrecMax	520	40	52	61	40	60	88	106	126	90	108	61	38	(16)
(17)		PrecMin	277	3	5	3	4	2	5	16	12	8	11	10	7	(17)
(18)		PrecStd	67	10	11	12	10	15	23	24	29	21	20	12	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.6	5.9	12.4	22.5	26.2	28.3	26.0	18.4	11.3	5.9	4.0	(19)
(20)			MCWB	1.1	1.8	2.5	7.0	12.8	16.1	17.5	17.6	13.4	9.3	5.0	2.4	(20)
(21)		2%	DB	0.6	1.3	3.2	8.9	17.9	22.5	25.7	23.1	16.2	9.6	3.2	2.1	(21)
(22)			MCWB	-0.2	0.1	0.9	4.6	10.4	13.9	16.5	16.0	12.1	7.9	2.4	1.1	(22)
(23)		5%	DB	-1.0	0.0	1.7	6.8	14.3	19.9	23.4	20.4	14.5	8.1	2.1	1.0	(23)
(24)			MCWB	-1.7	-1.0	-0.1	3.3	8.5	12.7	15.8	14.9	11.3	6.8	1.4	0.2	(24)
(25)		10%	DB	-2.6	-1.6	0.3	5.0	11.8	17.6	21.2	18.1	12.9	6.4	1.1	-0.2	(25)
(26)			MCWB	-3.2	-2.4	-1.2	2.0	7.3	11.7	15.1	13.9	10.5	5.3	0.6	-1.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	2.1	2.7	7.3	13.2	17.3	19.1	18.7	14.2	9.8	4.9	2.5	(27)
(28)			MCDWB	2.3	3.6	5.4	11.8	20.7	23.0	24.4	24.4	17.3	11.1	5.7	3.5	(28)
(29)		2%	WB	-0.2	0.2	1.2	5.0	11.2	15.2	18.0	17.0	12.9	8.0	2.6	1.2	(29)
(30)			MCDWB	0.4	1.2	2.9	8.1	16.9	20.8	23.2	21.7	15.3	9.1	3.0	2.0	(30)
(31)		5%	WB	-1.8	-1.0	0.0	3.6	9.1	13.7	16.9	15.5	11.7	6.8	1.4	0.3	(31)
(32)			MCDWB	-1.1	-0.1	1.4	6.4	13.8	18.1	21.7	19.2	14.0	8.0	1.9	1.0	(32)
(33)		10%	WB	-3.3	-2.4	-1.1	2.4	7.6	12.4	15.7	14.3	10.7	5.3	0.6	-1.1	(33)
(34)			MCDWB	-2.6	-1.6	0.3	4.5	11.2	17.0	20.0	17.6	12.6	6.3	1.0	-0.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.1	9.7	8.9	7.9	8.0	8.2	8.1	7.0	4.9	6.1	7.2	(35)
(36)			MCDBR	6.8	5.5	7.7	9.7	12.8	11.9	12.1	12.0	9.5	5.4	4.6	5.7	(36)
(37)			MCWBR	6.3	4.9	6.1	6.2	6.8	5.3	5.2	6.0	5.9	4.4	4.3	5.2	(37)
(38)		5% WB	MCDWR	6.9	5.6	6.8	8.8	11.8	10.2	10.3	10.4	8.1	5.1	4.1	5.7	(38)
(39)			MCWBR	6.4	5.1	5.7	5.8	6.9	5.2	5.2	5.7	5.4	4.3	3.9	5.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.194	0.248	0.290	0.321	0.342	0.343	0.355	0.337	0.306	0.267	0.354	N/A	(40)	
(41)		taud	1.917	2.069	2.153	2.219	2.406	2.477	2.465	2.492	2.547	2.284	2.732	N/A	(41)	
(42)		Ebn at Noon	64	521	728	818	848	858	832	808	746	513	111	0	(42)	
(43)		Edh at Noon	7	46	80	101	95	91	90	78	57	37	11	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.03	0.46	1.94	3.77	4.55	4.86	4.44	3.15	1.66	0.53	0.08	0.00	(44)	
(45)		RadStd	0.00	0.04	0.11	0.22	0.47	0.42	0.49	0.29	0.16	0.09	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page